DUAL-AXIS SOLAR TRACKER

FINAL BUILD BLUEPRINT • UPDATED MDF VERSION • EXPLODED VIEW • CUTTING LIST • PIN MAP

Revision: 23 September 2026

Fabrication authority: Use the written dimensions below. Measure actual component bodies before drilling or cutting. This revision supersedes the earlier plywood/base/deck dimensions.

1. FINAL DESIGN — FOUR LEVELS

Level 0: 20 mm MDF enclosure. Arduino UNO, breadboard/electronics, 4-channel relay, buck-boost, charging hardware, Samsung battery and voltage sensor inside. LCD, USB output, charging connector and master switch accessible outside.
Level 1: 20 mm MDF rotating azimuth deck, **140 mm corner-to-corner octagon**. Current azimuth MG90S is on D11 and directly drives the upper assembly.
Level 2: 3 mm acrylic U-frame. Current elevation MG90S is on D12 and directly drives elevation.
Level 3: 150 × 150 mm panel backing carrying four 70 × 70 mm panels, three LDRs, two-flap acrylic cover and D6 SG90 wiper.
Not included: INA219, dust sensor, dummy LDRs, slip ring and LDR multiplexer.

2. CURRENT BUILD CHECKPOINTS

Item	Status
Master switch	WORKING — tested successfully
Azimuth MG90S	TESTED — moves
Elevation MG90S	TESTED — moves
MDF enclosure	20 mm MDF; 200 × 200 × 100 mm external target
Azimuth deck	Updated to 140 mm corner-to-corner octagon
Tracking logic	Azimuth/elevation corrections operate independently
Servo power	Separate servo supplies with common Arduino ground

3. FINAL CUTTING LIST — 20 mm MDF

Part	Qty	Size (mm)	Notes
Base floor	1	200 × 200	Main enclosure base
Long walls	2	200 × 100	20 mm MDF stock
Short walls	2	160 × 100	Between long walls: 200 – 20 – 20
Fixed lid	1	200 × 200	Top/lid; verify actual fit before cutting
Battery-bay lid	1	≈70 × 70	Removable / screwed
Azimuth deck	1	140 mm octagon	140 mm corner-to-corner; ≈53.6 mm side
Azimuth servo block	1	≈30 × 25 × 20	D11 support; measure actual servo body

4. 3 mm ACRYLIC

Part	Qty	Size (mm)	Notes
Elevation side arms	2	90 × 30	Pivot holes
Elevation base strip	1	120 × 30	U-frame base
Elevation servo tab	1	25 × 20	D12 mount
Panel backing	1	150 × 150	Four 70 × 70 mm panels
Top LDR mount	1	≈40 × 15	TL + TR
Bottom LDR tab	1	≈20 × 15	Bottom LDR
Cover flaps	2	75 × 150	3 mm acrylic
Cover hinge reinforcement	2	150 × 10	Hinge mounting

5. WIPER

Part	Specification	Notes
Wiper arm	≈140 mm sweep	Across panel face
Blade	Soft rubber door-seal strip	Gentle contact
Pivot bracket	Sized to actual SG90	At one edge of 150 × 150 backing
Servo	SG90 on D6	Automated 0° → 160° → 0° dawn sweep

6. BASE + BATTERY + FRONT PANEL

Feature	Requirement
Battery	Samsung EB-BG360CBN measured approximately 50 × 55 × 10 mm
Battery bay	Approximately 65 × 70 mm footprint; ≈20 mm internal depth for clearance/inspection
LCD	16 × 2 I²C LCD; visible from front; SDA=A4, SCL=A5
USB output	External connector
Charging connector	Expose connector; keep charging module inside
Master switch	3-pin slide switch; center/common + one outer pin; third unused; rectangular slot

7. FINAL SERVO / SENSOR PIN MAP

Component	Hardware pin	Function
Azimuth MG90S	D11	Rotates upper assembly
Elevation MG90S	D12	Tilts panel frame
Cover SG90	D5	Clamshell cover
Wiper SG90	D6	Dawn sweep
LDR TL	A0	Light sensing
LDR TR	A1	Light sensing
LDR Bottom	A2	Light sensing
Voltage sensor	A3	Battery voltage
LCD SDA	A4	I²C data
LCD SCL	A5	I²C clock
Relay CH1	D4	USB charge control
Relay CH2–4	D2/D3/spare	Unused

8. TRACKING LOGIC

Three-LDR tracking uses TL versus TR for azimuth and the average of TL/TR versus Bottom for elevation. Both axes are allowed to correct independently; elevation is no longer blocked simply because azimuth moved.

Current settings: Azimuth 10–170° • Elevation 10–90° • LDR tolerance 50 • Servo step 1° • Move delay 60 ms • Night threshold 100 • Cover 0°/180° • Wiper 0°/160°.

9. ASSEMBLY ORDER / IMMEDIATE ACTIONS

#	Action	Verification
1	Measure actual four panel footprints.	Confirm 140 × 140 mm array fits 150 × 150 backing.
2	Cut/assemble 20 mm MDF base and walls.	Confirm 200 × 200 × 100 mm outside enclosure target.
3	Mark front cutouts.	Measure actual LCD, USB, charging connector and switch.
4	Build battery bay.	≈65 × 70 mm footprint; ≈20 mm depth; accessible.
5	Build 140 mm octagonal azimuth deck and D11 mount.	Free azimuth movement; direct drive.
6	Build acrylic U-frame and D12 mount.	No binding through elevation range.
7	Mount panels and LDRs.	TL/TR top; bottom centered; divider between top LDRs.
8	Build cover.	Both 75 × 150 mm flaps clear panels and move together.
9	Build wiper.	≈140 mm sweep; soft blade; no scraping.
10	Install electronics.	Keep moving parts and wiring separated.
11	Run final firmware test.	Verify D11/D12/D5/D6 and A0–A3.

10. FINAL PRE-CUT CHECK

■ Panel array measured ■ Battery ≈50×55×10 mm ■ Bay ≈65×70×20 mm ■ Servo bodies/horns measured
■ Front cutouts measured from hardware ■ Wiper reaches ≈140 mm ■ 140 mm azimuth octagon marked

11. MDF CUTTING REFERENCE

Enclosure: Base 200 × 200 mm • Long walls 200 × 100 mm ×2 • Short walls 160 × 100 mm ×2 • 20 mm MDF.
Azimuth: 140 mm corner-to-corner regular octagon • radius 70 mm • side ≈53.6 mm (mark ≈54 mm).

IMPORTANT: Do not permanently glue servo mounts or other moving components until the mechanical clearances and cable routing have been checked. The master switch has already been tested and works correctly.